Model Predictive Control — Spring 2026

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Exercise 10

Robust MPC 2

Exercise 1 **Tube-based MPC**

Consider the scalar linear discrete time system

$$x(k+1) = 1.1x(k) + u(k) + w(k),$$

where the state x and the input u are constrained as follows

$$\mathcal{X} = [-2, 2] \qquad \mathcal{U} = [-0.8, 0.8]$$

and the disturbance $w \in \mathcal{W} = [-0.2, 0.1]$. Design a tube-based MPC controller for this system.

1. Compute the minimal robust invariant set $\mathcal{E}$ for this system with the control law $u(k) = Kx(k)$, for $K = -0.3$

   Hint : $[a, b] \oplus [c, d] = [a + c, b + d]$ and $\sum_{i=0}^{\infty} x^i = \frac{1}{1-x}$ if $|x| < 1$
2. Compute the tightened constraints $\mathcal{X} \ominus \mathcal{E}$ and $\mathcal{U} \ominus K\mathcal{E}$

   Hint : $[a, b] \ominus [c, d] = [a - c, b - d]$
3. Verify that the set $\mathcal{X}_f = [-1, 1.5]$ if used as a terminal constraint for the tube-based MPC scheme will result in a recursively feasible controller for the terminal control law $u(k) = -0.3x(k)$

Exercise 2 **Nominal MPC with disturbances**

Consider the scalar linear discrete time system

$$x(k+1) = 0.9x(k) + u(k) + w(k),$$

where the state and the input constraints are given by

$$\mathcal{X} = [-3, 3] \qquad \mathcal{U} = [-0.05, 0.05],$$

the disturbance $w \in \mathcal{W} = [-0.3, 0.3]$, and the initial condition is $x(0) = 3$.

1. Design a nominal MPC controller with horizon length $N = 2$, $Q = 1$ and $R = 1$
2. Run a simulation for 100 time step starting from $x(0)$. Does the state converge to a neighborhood of the origin?
3. What happens if we choose $w \in \mathcal{W} = [-1, 1]$?